

RBF-FD Solver: Least Squares Method for Operator Reproduction

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Contents

1	Problem Setting	2
2	Functional Analysis Aspect	3
3	RBF-FD Method	5
3.1	Local Approximation setup	5
3.2	Polynomial Augmentation	5
3.3	Consistency Error Analysis	7
3.4	Growth Function	8
4	Local Least Square Weights RBF-FD	8
4.1	Weight Constraint Setup	8
4.2	Consistency Conditions for Finite Spaces and Interpolation Results	9
4.3	Consistency Results for Function Approximations	10
5	Appendix	12

1 Problem Setting

Following the standard RBF-FD methods, let $u \in \mathcal{F} = W^{k,p}$ be a solution for a PDE in form of:

$$-Lu = -\nabla \cdot (A \nabla u) = f \quad \text{in } \Omega, \quad (1)$$

Let $1 < p < \infty$, $A \in W^{1,\infty}(\bar{\Omega}; \mathbb{R}^{d \times d}) : \bar{\Omega} \rightarrow \mathbb{R}^{d \times d}$ is an symmetric positive definite anisotropic tensor, where

$$\zeta |\xi|^2 \leq \mathbf{A}(\mathbf{x}) \xi \cdot \xi \leq \mathbf{Z} |\xi|^2, \quad \mathbf{0} < \zeta \leq \mathbf{Z} \quad (2)$$

Assume $\Omega \subset \mathbb{R}^d$ and $f \in W^{k-t,p}(\Omega)$ is the forcing term acting in the domain Ω . For 2^{nd} order elliptic operators, we set $t = 2$. The problem is subject to Neumann and Dirichlet Boundary Conditions partitioned within $\partial\Omega$ of the form:

$$u = g_D \quad \text{on } \Gamma_D \subset \partial\Omega, \quad (3)$$

$$n \cdot A \nabla u = g_N \quad \text{on } \Gamma_N \subset \partial\Omega, \quad (4)$$

For simplicity, we assume Ω be a bounded $C^{1,1}(\partial\Omega)$ domain, with $\partial\Omega = \Gamma_D \cup \Gamma_N$ and $\Gamma_D \cap \Gamma_N = \emptyset$. Furthermore, let $g_D \in W^{k-1/p,p}(\partial\Omega)$, $g_N \in W^{k-1-1/p,p}(\partial\Omega)$. By *Agmon-Douglis-Nirenberg* Theorem [2], we have the following estimate for u :

$$\|u\|_{W^{k,p}(\Omega)} \leq C \left(\|Lu\|_{W^{k-2,p}(\Omega)} + \|g_D\|_{W^{k-1/p,p}(\partial\Omega)} + \|g_N\|_{W^{k-1-1/p,p}(\partial\Omega)} + \|u\|_{L^p(\Omega)} \right). \quad (5)$$

In the case that we work on a Hilbert space, we then have for $p = 2$ ($W^{k,2} := H^k$) the following estimate for u :

$$\|u\|_{H^k(\Omega)} \leq C \left(\|Lu\|_{H^{k-2}(\Omega)} + \|g_D\|_{H^{k-\frac{1}{2}}(\partial\Omega)} + \|g_N\|_{H^{k-\frac{3}{2}}(\partial\Omega)} + \|u\|_{L^2(\Omega)} \right).$$

If we want to study cases where $(Lu)(\mathbf{x})$ is known pointwise, we can formulate the setup by having $f \in W^{s-t,p}(\Omega)$ and, by Morrey Embedding,

$$W^{s-t,p} \hookrightarrow C^{\mu,\gamma}, \quad \text{with } \mu := \left\lfloor s - t - \frac{d}{p} \right\rfloor, \quad \gamma := s - t - \mu - \frac{d}{p}$$

In other words, $s > t + \frac{d}{p}$. Suppose $p = 2$ and $t = 2$, then $s > 2 + \frac{d}{2}$. Given an operator $L : H^s(\Omega) \rightarrow H^{s-2}(\Omega)$, we define an error functional of the form:

$$\xi_{L,x} : H^s(\Omega) \rightarrow \mathbb{R}, \quad \xi_{L,x} \in (H^s)^*(\Omega)$$

Then by Riesz Representation Theorem, there exists a $v_{L,x} \in H^s(\Omega)$ such that $\xi_{L,x}(u) = \langle u, v_{L,x} \rangle_{H^s(\Omega)}$, and $\|\xi_{L,x}\|_{(H^s(\Omega))^*} = \|v_{L,x}\|_{H^s(\Omega)}$

2 Functional Analysis Aspect

We examine the structural framework that governs the RBF-FD method. Let $\mathcal{N}(\Omega) := W^{s,p}(\Omega)$ be the native space in which solutions are sought. We later assume that $\mathcal{N}(\Omega)$ is a Hilbert space (this holds, for instance, when $\mathcal{N}(\Omega)$ is the native space of a symmetric positive definite kernel). Let $L : \mathcal{N}(\Omega) \rightarrow W^{s-t,p}(\Omega)$ be a bounded linear operator.

For $n \in \mathbb{N}$, let $\theta_1, \dots, \theta_n \in \mathcal{N}(\Omega)^*$ be bounded linear functionals (for example, point evaluations $\theta_i = \delta_{\mathbf{x}_i}$), and let $w_1, \dots, w_n \in W^{s-t,p}(\Omega)$ be weight functions. Define $L_h : \mathcal{N}(\Omega) \rightarrow W^{s-t,p}(\Omega)$ by

$$L_w u := \sum_{i=1}^n w_i \theta_i(u). \quad (6)$$

This operator is linear and bounded, since

$$\|L_w u\|_{W^{s-t,p}(\Omega)} \leq \sum_{i=1}^n \|w_i\|_{W^{s-t,p}(\Omega)} \|\theta_i\|_{\mathcal{N}(\Omega)^*} \|u\|_{\mathcal{N}(\Omega)}.$$

The goal is for L_w to approximate L .

We further assume that there is a linear subspace $\mathcal{V}_h(\Omega) \subseteq \mathcal{N}(\Omega)$ on which L_w reproduces L exactly, i.e., $Lu = L_h u$ for all $u \in \mathcal{V}_h(\Omega)$. We define the error operator $\xi_{w,L} \in \mathcal{L}(\mathcal{N}(\Omega), W^{s-t,p}(\Omega))$ by

$$\xi_{w,L}(u) := Lu - L_w u.$$

It is linear and bounded because L and L_w are, and it vanishes on $\mathcal{V}_h(\Omega)$.

Let $1 < p < \infty$ and let q satisfy $\frac{1}{p} + \frac{1}{q} = 1$. For $g \in W^{s-t,q}(\Omega)$ define the linear functional l_g on $W^{s-t,p}(\Omega)$ by

$$l_g(v) := \sum_{|\alpha| \leq k} \int_{\Omega} D^{\alpha} v D^{\alpha} g \, d\mathbf{x}.$$

With $\|v\|_{W^{s-t,p}(\Omega)} = (\sum_{|\alpha| \leq k} \|D^{\alpha} v\|_{L^p(\Omega)}^p)^{1/p}$, the Hölder inequality (applied to the integrals and then to the finite sum over α) gives

$$|l_g(v)| \leq \|v\|_{W^{s-t,p}(\Omega)} \|g\|_{W^{k,q}(\Omega)}.$$

Set $\xi_{w,L,g} := l_g \circ \xi_{w,L}$. It is linear, and for every $u \in \mathcal{N}(\Omega)$

$$\begin{aligned} |\xi_{w,L,g}(u)| &\leq \|\xi_{w,L}(u)\|_{W^{s-t,p}(\Omega)} \|g\|_{W^{s-t,q}(\Omega)} \\ &\leq \|\xi_{w,L}\|_{\mathcal{L}(\mathcal{N}, W^{s-t,p})} \|g\|_{W^{s-t,q}(\Omega)} \|u\|_{\mathcal{N}(\Omega)}. \end{aligned}$$

Thus $\xi_{w,L,g}$ is a bounded functional, $\xi_{w,L,g} \in \mathcal{N}(\Omega)^*$. Assuming now that $\mathcal{N}(\Omega)$ is a Hilbert space, the Riesz Representation Theorem gives a unique $v_{w,L,g} \in \mathcal{N}(\Omega)$ such that $\xi_{w,L,g}(u) = \langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}$ for all $u \in \mathcal{N}(\Omega)$. Moreover,

$$\begin{aligned} \|\xi_{w,L,g}\|_{\mathcal{N}(\Omega)^*} &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ \|u\|_{\mathcal{N}(\Omega)} \leq 1}} |\xi_{w,L,g}(u)| \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\xi_{w,L,g}(u)|}{\|u\|_{\mathcal{N}(\Omega)}} \\ &= \sup_{\substack{u \in \mathcal{N}(\Omega) \\ u \neq 0}} \frac{|\langle u, v_{w,L,g} \rangle_{\mathcal{N}(\Omega)}|}{\|u\|_{\mathcal{N}(\Omega)}} = \|v_{w,L,g}\|_{\mathcal{N}(\Omega)}, \end{aligned}$$

where the last equality is the Cauchy–Schwarz inequality, attained at $u = v_{w,L,g}$ (the case $v_{w,L,g} = 0$ is trivial). Finally, for $u \in \mathcal{N}(\Omega)$, we have the following:

$$|\xi_{w,L,g}(u)| \leq \|v_{w,L,g}\|_{\mathcal{N}(\Omega)} \|u\|_{\mathcal{N}(\Omega)}.$$

For pointwise estimates, point values of $\xi_{w,L}(u)$ must be meaningful, and elements of $W^{s-t,p}(\Omega)$ are in general only equivalence classes. Consider the case $s > t + \frac{d}{p}$, where every element of $W^{s-t,p}(\Omega) \hookrightarrow C^{\mu,\gamma}(\bar{\Omega})$ is also Hölder continuous with $\gamma = s - t - \mu - \frac{d}{p}$. Then for every $\mathbf{x} \in \Omega$

$$|\xi_{w,L}(u)(\mathbf{x})| \leq \|\xi_{w,L}(u)\|_{C^{\mu,\gamma}(\bar{\Omega})} \leq C_{\mathcal{H}} \|\xi_{w,L}(u)\|_{W^{s-t,p}(\Omega)} \leq \|\xi_{w,L}\|_{\mathcal{L}(\mathcal{N}, W^{s-t,p})} \|u\|_{\mathcal{N}(\Omega)},$$

so $\xi_{w,L,\mathbf{x}} := \delta_{\mathbf{x}} \circ \xi_{w,L}$ is a bounded linear functional, $\xi_{w,L,\mathbf{x}} \in \mathcal{N}(\Omega)^*$. Consider $p = 2$ and by the Riesz Representation Theorem, there exists $v_{w,L,\mathbf{x}} \in \mathcal{N}(\Omega)$ such that $\xi_{w,L,\mathbf{x}}(u) = \langle u, v_{w,L,\mathbf{x}} \rangle_{\mathcal{N}(\Omega)}$, and, as above, $\|\xi_{w,L,\mathbf{x}}\|_{\mathcal{N}(\Omega)^*} = \|v_{w,L,\mathbf{x}}\|_{\mathcal{N}(\Omega)}$. Finally, we have the pointwise error estimate on functions u by the approximation operator L_h in the form

$$|(Lu)(\mathbf{x}) - (L_h u)(\mathbf{x})| = |\xi_{w,L,\mathbf{x}}(u)| \leq \|v_{w,L,\mathbf{x}}\|_{\mathcal{N}(\Omega)} \|u\|_{\mathcal{N}(\Omega)}. \quad (7)$$

3 RBF-FD Method

3.1 Local Approximation setup

To obtain an approximation of the operator L at a point $\mathbf{z} \in \Omega$, we approximate u by a function $\hat{u}$ on a neighborhood $\Omega_{\mathbf{z}} \subseteq \Omega$ of $\mathbf{z}$, so that $Lu(\mathbf{z}) \approx L\hat{u}(\mathbf{z})$. Let $Y = \{\mathbf{y}_1, \dots, \mathbf{y}_m\} \subseteq \Omega_{\mathbf{z}}$ be a set of centers, let $\phi_i : \Omega_{\mathbf{z}} \rightarrow \mathbb{R}$, $i = 1, \dots, m$, be basis functions, and define the vector space $V_Y = \text{span}\{\phi_1, \dots, \phi_m\}$. Let $X = \{\mathbf{x}_1, \dots, \mathbf{x}_n\} \subseteq \Omega_{\mathbf{z}}$ be the set of stencil nodes for $\mathbf{z}$. For the classical approximation scheme, let $m = n$. We define $\Phi \in \mathbb{R}^{n \times m}$ by $\Phi_{ij} = \phi_j(\mathbf{x}_i)$ and assume that Φ is invertible (for example, $\mathbf{y}_i = \mathbf{x}_i$ and a strictly positive definite kernel). The role of these basis functions will be to approximate the operator L . We seek $\hat{u} \in V_Y$ of the form

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n \lambda_i \phi_i(\mathbf{x}) = \phi(\mathbf{x})^T \boldsymbol{\lambda}, \quad \phi(\mathbf{x}) := [\phi_1(\mathbf{x}), \dots, \phi_n(\mathbf{x})]^T, \quad (8)$$

that interpolates u on the stencil, $\hat{u}(\mathbf{x}_i) = u(\mathbf{x}_i)$ for $i = 1, \dots, n$. This leads to the form $\Phi \boldsymbol{\lambda} = \mathbf{u}$ for $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. We can solve for $\boldsymbol{\lambda}$ to obtain:

$$\hat{u}(\mathbf{x}) = \sum_{i=1}^n (\Phi^{-1} \mathbf{u})_i \phi_i(\mathbf{x}) = \phi(\mathbf{x})^T \Phi^{-1} \mathbf{u}.$$

Applying L (which acts on the argument $\mathbf{x}$ of ϕ) and evaluating at $\mathbf{z}$, the approximation of Lu at $\mathbf{z}$ is

$$Lu(\mathbf{z}) \approx L\hat{u}(\mathbf{z}) = (L\phi)(\mathbf{z})^T \boldsymbol{\lambda} = (L\phi)(\mathbf{z})^T \Phi^{-1} \mathbf{u} =: \mathbf{w}^{L,X}(\mathbf{z})^T \mathbf{u}, \quad (9)$$

where $(L\phi)(\mathbf{z}) := [(L\phi_1)(\mathbf{z}), \dots, (L\phi_n)(\mathbf{z})]^T$ and the weights $\mathbf{w}^{L,X}(\mathbf{z}) \in \mathbb{R}^n$ form a column vector.

Thus, approximating the operator reduces to a local linear solve for the weights. Transposing the definition of $\mathbf{w}^{L,X}(\mathbf{z})$ gives

$$\Phi^T \mathbf{w}^{L,X}(\mathbf{z}) = (L\phi)(\mathbf{z}).$$

The method of undetermined coefficients on the basis $\{\phi_i\}$ gives the same weights: requiring exactness on each basis function, $\sum_{i=1}^n w_i \phi_j(\mathbf{x}_i) = (L\phi_j)(\mathbf{z})$ for $j = 1, \dots, n$, is exactly the system above. Finally, we let the basis of V_Y be constituted by radial basis functions, $\phi_i(\mathbf{x}) := \varphi(\|\mathbf{x} - \mathbf{y}_i\|)$, with a radial function $\varphi : [0, \infty) \rightarrow \mathbb{R}$ and the centers $\{\mathbf{y}_i\}_{i=1}^m \subseteq \Omega_{\mathbf{z}}$ introduced above. Thus we have defined the Radial Basis Function-generated Finite Difference method (RBF-FD).

Remark. For $\mathbf{x} \in \Omega_{\mathbf{z}}$ let $w_i(\mathbf{x}) := (\mathbf{w}^{L,X}(\mathbf{x}))_i$ be the weights computed with the fixed stencil X , and take $\theta_i = \delta_{\mathbf{x}_i}$. Then $L_h u = \sum_{i=1}^n w_i \theta_i(u) = L\hat{u}$ on $\Omega_{\mathbf{z}}$, that is, $L_h = L \circ I_X$ where $I_X u := \hat{u}$ is the interpolation operator. If $u \in V_Y$, then $I_X u = u$ and hence $L_h u = Lu$. So the exactness assumption of the previous section holds with $\mathcal{V}_h = V_Y$ (working on $\Omega_{\mathbf{z}}$ in place of Ω and assuming $V_Y \subseteq \mathcal{N}$).

3.2 Polynomial Augmentation

For conditionally positive definite kernels, we augment them through the form:

$$\hat{u}(\mathbf{x}) = \sum_{j=1}^n \lambda_j \phi_j(\mathbf{x}) + \sum_{j=1}^1 \beta_j \mathbf{p}_j(\mathbf{x}) \quad (10)$$

$$\sum_{i=1}^n \lambda_i p_j(\mathbf{x}_i) = 0, \quad \forall j = 1, \dots, n \quad (11)$$

We can generate a Gram matrix equation as follows:

$$\begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \boldsymbol{\lambda} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Hence, we can approximate the operator as below:

$$(L\hat{\mathbf{u}})(\mathbf{x}) = \begin{bmatrix} (\mathbf{L}\phi)(\mathbf{x}) \\ (\mathbf{L}\mathbf{p})(\mathbf{x}) \end{bmatrix}^T \begin{bmatrix} \mathbf{\Phi} & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} := \begin{bmatrix} \mathbf{w}(\mathbf{x}) \\ \boldsymbol{\nu}(\mathbf{x}) \end{bmatrix}^T \begin{bmatrix} \hat{\mathbf{u}} \\ \mathbf{0} \end{bmatrix} = \mathbf{w}^T(\mathbf{x})\hat{\mathbf{u}} \quad (12)$$

Thus, we get the local augmented operator reproduction scheme:

$$\begin{bmatrix} (\mathbf{L}\phi)(\mathbf{x}) \\ (\mathbf{L}\mathbf{p})(\mathbf{x}) \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w}(\mathbf{x}) \\ \boldsymbol{\nu}(\mathbf{x}) \end{bmatrix} \quad (13)$$

The polynomial constraint setup can be viewed as a Lagrange multiplier to a minimization scheme for the weights. By [6], given the basis for V_Y , $(\mathbf{L}\mathbf{p})(\mathbf{x}) \in \text{Range}(\mathbf{P}^T) \iff \text{Eq. (13)}$ is uniquely solvable, for any set X . Additionally, by [6], Eq. (12) hold $\iff$ Eq. (13) holds. Note that depending on the the basis V_Y , we get the degree of $p \in \Pi_{k-1}^d$. Below are common RBFs and their respective cpd orders:

Table 1: Common radial basis functions and their minimal CPD order.

RBF	Radial function $\phi(r)$	CPD order k
Gaussian	$\exp(-\varepsilon^2 r^2)$	0
Inverse multiquadric	$(r^2 + \varepsilon^2)^{-1/2}$	0
Multiquadric	$(r^2 + \varepsilon^2)^{1/2}$	1
Linear	r	1
Cubic	r^3	2
Thin-plate spline	$r^2 \log r$	2
Quintic	r^5	3

This order provides us both, the degree of the polynomials $k - 1$ and the number of terms:

$$l = \binom{d + k - 1}{k - 1}$$

As shown in [8], a polynomial augmentation can be added with a larger degree than required for the RBF, acting as a correction term while inheriting the non-oscillatory behavior of the RBF at the boundary $\partial\Omega$.

3.3 Consistency Error Analysis

We begin by defining the native space of solutions for (1) with (4). Suppose $\mathcal{F}_K = H^s(\Omega)$ with pointwise bounded evaluations. This defines an RKHS for functions $u \in H^s(\Omega)$, $Lu \in H^{s-2}(\Omega)$. We can also define a *Kernel* $K : \Omega \times \Omega \rightarrow \mathbb{R}$ such that:

$$\begin{aligned} f(x) &= \langle f, K(\cdot, x) \rangle_{\mathcal{F}_K} \\ (Lf)(x) &= \langle f, L_x K(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_K} \\ \mathbf{w}^T(\mathbf{x}) \mathbf{f} &= \langle f, \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_K} \\ \xi_{w,L,x}(f) &:= Lf(x) - \mathbf{w}^T(\mathbf{x}) \mathbf{f} \end{aligned}$$

L_x indicates applying the operator acted on the variable $\mathbf{x}$ in the kernel. We define $\mathbf{k}(\mathbf{z}) = [k(\mathbf{z}, \mathbf{x}_1), \dots, k(\mathbf{z}, \mathbf{x}_n)]^T$. Then $\xi_{w,L,x}(f) = \langle f, v_{w,L,x} \rangle$, with $v_{w,L,x} = L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot)$. By assuming point-wise error bounds, $\|\xi_{w,L,x}\|_{\mathcal{F}_K^*}$, with error estimate $\|\xi_{w,L,x}\|_{\mathcal{F}_K^*} = \|v_{w,K,L,X,x}\|_{\mathcal{F}_K}$. By [5], the functional is continuous on $\mathcal{F}_K$ if $L_x K$ exists and both, polynomial consistency and operator exactness are guaranteed. Finally, We define the power function:

$$\mathcal{P}_{K,L,X}^2(x) = \min \left\{ \|v_{w,K,L,X,x}\|_{\mathcal{F}_K}^2 : \mathbf{w} \in \mathbb{R}^n, \xi_{w,L,x} p = 0 \text{ for } p \in \Pi_{k-1}^d \right\} \quad (14)$$

The RKHS norm can be simplified as

$$\begin{aligned} \|v_{w,K,L,X,x}\|_{\mathcal{F}_K}^2 &= \langle v_{w,K,L,X,x}, v_{w,K,L,X,x} \rangle_{\mathcal{F}_K} \\ &= \xi_{w,L,y} \xi_{w,L,z} K(y, z)|_{y=z=x} \\ &= \langle L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot), L_x K(\cdot, \mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_K} \\ &= \langle L_x K(\cdot, \mathbf{x}), L_x K(\cdot, \mathbf{x}) \rangle_{\mathcal{F}_K} - 2 \langle L_x K(\cdot, \mathbf{x}), \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_K} + \langle \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot), \mathbf{w}^T(\mathbf{x}) \mathbf{k}(\cdot) \rangle_{\mathcal{F}_K} \\ &= L_y L_z K(y, z)|_{y=z=x} - 2 \sum_{i=1}^n w_i(\mathbf{x}) L_x K(\mathbf{x}, \mathbf{x}_i) + \sum_{i,j=1}^n w_i(\mathbf{x}) w_j(\mathbf{x}) K(\mathbf{x}_i, \mathbf{x}_j) \\ &= L_y L_z K(y, z)|_{y=z=x} - \sum_{i=1}^n w_i(\mathbf{x}) L_x K(\mathbf{x}, \mathbf{x}_i) - \sum_{i=1}^l v_i L p_i(\mathbf{x}) \\ & \quad (*\text{Need to check this line since [6] cancels the third term.}*) \end{aligned}$$

The last two line are carried out by Eq (13).

Now we have an error estimate for functions in $\mathcal{F}_K$ of the form:

$$\begin{aligned} |\xi_{w,L,x}(f)| &\leq \|\xi_{w,L,x}\|_{\mathcal{F}_K^*} \|f\|_{\mathcal{F}_K} \\ |\xi_{w^*,L,x}(f)| &\leq \inf_{\substack{\mathbf{w} \in \mathbb{R}^n \\ \xi(\Pi_{k-1}^d)=0}} \|\xi_{w,L,x}\|_{\mathcal{F}_K^*} \|f\|_{\mathcal{F}_K} \\ |\xi_{w^*,L,x}(f)| &\leq \inf_{\substack{\mathbf{w} \in \mathbb{R}^n \\ \xi(\Pi_{k-1}^d)=0}} \|v_{w,K,L,X,x}\|_{\mathcal{F}_K} \|f\|_{\mathcal{F}_K} \\ |\xi_{w^*,L,x}(f)| &\leq \mathcal{P}_{K,L,X}(\mathbf{x}) \|f\|_{\mathcal{F}_K} \end{aligned}$$

3.4 Growth Function

To define the consistency order, we define the following stencil as the stencil set.

$$S_{\mathbf{z},X} := \bigcup_{i=1}^N [\mathbf{z}, \mathbf{x}_i]$$

Then for $U : S_{\mathbf{z},X} \times S_{\mathbf{z},X} \rightarrow \mathbb{R}$, we define a binary Höder continuity semi-norm in the form:

$$|U|_{\mathbf{z},X,\gamma} := \sup_{\substack{\mathbf{x}, \mathbf{y} \in S_{\mathbf{z},X} \\ \mathbf{x} \neq \mathbf{z}, \mathbf{y} \neq \mathbf{z}}} \frac{|U(\mathbf{x}, \mathbf{y}) - U(\mathbf{x}, \mathbf{z}) - U(\mathbf{z}, \mathbf{y}) + U(\mathbf{z}, \mathbf{z})|}{\|\mathbf{x} - \mathbf{z}\|_2^\gamma \|\mathbf{y} - \mathbf{z}\|_2^\gamma}, 0 < \gamma, \leq 1$$

Then, for $r \in \mathbb{N}$, we get

$$|U|_{\mathbf{z},X,r+\gamma} := \frac{1}{(\gamma+1)^2 \dots (\gamma+r)^2} \left(\sum_{|\alpha|=r} \sum_{|\beta|=r} \binom{r}{\alpha} \binom{r}{\beta} |\partial^{\alpha,\beta} U|_{\mathbf{z},X,\gamma}^2 \right)^{1/2},$$

with the following coefficient definitions: $\binom{r}{\alpha} = \frac{r!}{\alpha!}$, $\alpha! = \alpha_1! \dots \alpha_d!$.

Growth function is defined as follows:

$$\rho_{k-1,D}(\mathbf{z}, X, \mu) := \sup \left\{ Lp(\mathbf{z}) : p \in \Pi_{k-1}^d, |p(\mathbf{x}_j)| \leq \|\mathbf{x}_j - \mathbf{z}\|_2^\mu, j = 1, \dots, n \right\} \quad (15)$$

By [9], the dual form of ρ takes the form

$$\rho_{k-1,D}(\mathbf{z}, Z, \|\cdot\|) := \inf \left\{ \|\mathbf{w}\| : \mathbf{w} \in \mathbb{R}^n, \xi_{w,L,x} p = 0, p \in \Pi_{k-1}^d \right\}$$

4 Local Least Square Weights RBF-FD

4.1 Weight Constraint Setup

Rather than assuming a form for the solution, our goal is to approximate the operator through weights dual to u sampled at nodes. Suppose $m > n$ (for the case of $m < n$, refer to [1]), thus more center nodes than stencil nodes per stencil. Additionally, instead of assuming exactness of the form $L\phi_j(\mathbf{x}) = \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i)$ for $j = 1, \dots, m$ as in the classical RBF-FD, we define a residual function:

$$\begin{aligned} r_j(\mathbf{x}) &:= L\phi_j(\mathbf{x}) - \sum_{i=1}^n w_i(\mathbf{x})\phi_j(\mathbf{x}_i) \\ r(\mathbf{x}) &= L\phi(\mathbf{x}) - \Phi^T \mathbf{w}(\mathbf{x}), \quad \mathbf{w} : \mathbb{R}^n \rightarrow \mathbb{R}^n \text{ for } \mathbf{x} \in \Omega. \end{aligned}$$

Suppose $\{p_i\}_{i=1}^l$ is a set of polynomials augmenting the radial basis functions. Let $\Pi_k = \text{span} \{p_i\}_{i=1}^l$ with polynomial degree up to k . We want to reproduce the operator exactly on the polynomial basis:

$$\begin{aligned} Lp_j(\mathbf{x}) &:= \sum_{i=1}^n w_i(\mathbf{x})p_j(\mathbf{x}_i) \text{ for } j = 1, \dots, l \\ Lp(\mathbf{x}) &= \mathbf{P}^T \mathbf{w} \end{aligned}$$

where $\mathbf{P}_{ij} := p_j(\mathbf{x}_i) \in \mathbb{R}^{n \times l}$. Thus, we have a constrained least squares problem in the form:

$$\begin{cases} \min J(\mathbf{w}(\mathbf{x})) := \min \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|_2^2 & \text{for } \mathbf{x} \in \Omega \\ \text{subject to } \mathbf{\Phi}^T \mathbf{w} = L\mathbf{p} \end{cases} \quad (16)$$

We can solve this setup by defining the Lagrangian $\mathcal{L}$ for the problem as follows:

$$\begin{aligned} \mathcal{L}(\mathbf{w}, \boldsymbol{\nu}) &:= J(\mathbf{w}) + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p}(\mathbf{x}) \rangle \\ &= \frac{1}{2} \langle \mathbf{\Phi}^T \mathbf{w}, \mathbf{\Phi}^T \mathbf{w} \rangle - \langle \mathbf{\Phi}^T \mathbf{w}, L\boldsymbol{\phi} \rangle + \frac{1}{2} \langle L\boldsymbol{\phi}, L\boldsymbol{\phi} \rangle \\ &\quad + \langle \boldsymbol{\nu}, \mathbf{P}^T \mathbf{w} - L\mathbf{p}(\mathbf{x}) \rangle. \end{aligned}$$

4.2 Consistency Conditions for Finite Spaces and Interpolation Results

The resulting KKT equations for this system are then:

$$\begin{aligned} \nabla_{\mathbf{w}} \mathcal{L} &= 0 : \mathbf{\Phi} L\boldsymbol{\phi} = \mathbf{\Phi} \mathbf{\Phi}^T \mathbf{w} + \mathbf{P} \boldsymbol{\nu} \\ \nabla_{\boldsymbol{\nu}} \mathcal{L} &= 0 : \mathbf{P}^T \mathbf{w} = L\mathbf{p} \end{aligned}$$

The augmented matrix then takes the form:

$$\begin{bmatrix} \mathbf{\Phi} \mathbf{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{w} \\ \boldsymbol{\nu} \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi} L\boldsymbol{\phi} \\ L\mathbf{p} \end{bmatrix} \quad (17)$$

From [6], Equation (17) has a unique solution given $L\mathbf{p} \in \text{Range}(\mathbf{P}^T)$. The primary goal is to approximate $Lu(\mathbf{x}) \approx \mathbf{w}^T(\mathbf{x})\mathbf{u}$, with $\mathbf{u} = [u(\mathbf{x}_1), \dots, u(\mathbf{x}_n)]^T$. Define $L_h f(\mathbf{x}) := \mathbf{w}^T(\mathbf{x})\mathbf{f}$ for some function f . Thus, we have the following:

$$\begin{aligned} L_h f(\mathbf{x}) &= \mathbf{w}^T(\mathbf{x})\mathbf{f} \\ &= [\mathbf{w}^T \quad \boldsymbol{\nu}^T] \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= \left(\begin{bmatrix} \mathbf{\Phi} \mathbf{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{\Phi} L\boldsymbol{\phi} \\ L\mathbf{p} \end{bmatrix} \right)^T \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= [L\boldsymbol{\phi}^T \mathbf{\Phi}^T \quad L\mathbf{p}^T] \begin{bmatrix} \mathbf{\Phi} \mathbf{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix}^{-T} \begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} \\ &= [L\boldsymbol{\phi}^T \mathbf{\Phi}^T \quad L\mathbf{p}^T] \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix} \\ &= L\boldsymbol{\phi}^T(\mathbf{x}) (\mathbf{\Phi}^T \mathbf{q}) + L\mathbf{p}^T(\mathbf{x}) \boldsymbol{\alpha} \end{aligned}$$

Next, we can show that the coefficients $\mathbf{q}, \boldsymbol{\alpha}$ relate to the coefficients of u as follows:

$$\begin{bmatrix} \mathbf{f} \\ \mathbf{0} \end{bmatrix} = \begin{bmatrix} \mathbf{\Phi} \mathbf{\Phi}^T & \mathbf{P} \\ \mathbf{P}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \boldsymbol{\alpha} \end{bmatrix}$$

Define $V_h = \{\rho(x) : \rho(x) = \sum_{i=1}^m \lambda_i \phi_i(x) \text{ such that } \lambda \in \text{Range}(\mathbf{\Phi}^T)\} \subseteq V_Y$. Let $s \in V_h + \Pi_k$. Thus, $s(x) = \boldsymbol{\phi}^T(\mathbf{x})\mathbf{\Phi}^T \mathbf{q} + \mathbf{p}^T(\mathbf{x})\boldsymbol{\alpha}$ where $\mathbf{q} \in \mathbb{R}^n$ and $\boldsymbol{\alpha} \in \mathbb{R}^l$. Then:

$$\begin{aligned} \mathbf{s} &= s(X) = \mathbf{\Phi} \mathbf{\Phi}^T \mathbf{q} + \mathbf{P} \boldsymbol{\alpha} = f(X) = \mathbf{f} \\ \mathbf{P}^T \mathbf{q} &= \mathbf{0} \end{aligned}$$

Hence, we get interpolation of f at the node set X . Furthermore, from the KKT matrix we can show that $\Phi \mathbf{r} = \mathbf{P} \boldsymbol{\nu}$ which implies that $\Phi \mathbf{r} \in \text{range}(\mathbf{P}) = (\ker(\mathbf{P}^T))^\perp$. Thus, we get the relationship:

$$\mathbf{r}^T (\Phi^T \mathbf{z}) = \mathbf{0} \text{ for } \mathbf{P}^T \mathbf{z} = \mathbf{0} \quad (18)$$

Finally, we check the consistency error for $s \in V_h + \Pi_k$ with $\mathbf{P}^T \mathbf{q} = \mathbf{0}$.

$$\begin{aligned} Ls(x) - L_h s(x) &= L(\phi^T(x) \Phi^T \mathbf{q} + \mathbf{p}^T(x) \boldsymbol{\alpha}) - \mathbf{w}^T(x) s \\ &= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \boldsymbol{\alpha} - \mathbf{w}^T(x) (\Phi \Phi^T \mathbf{q} + \mathbf{P} \boldsymbol{\alpha}) \\ &= L\phi^T(x) \Phi^T \mathbf{q} + L\mathbf{p}^T(x) \boldsymbol{\alpha} - \mathbf{w}^T(x) \Phi \Phi^T \mathbf{q} - \mathbf{w}^T(x) \mathbf{P} \boldsymbol{\alpha} \\ &= (L\phi(x) - \Phi^T \mathbf{w}(x))^T \Phi^T \mathbf{q} + (L\mathbf{p}^T(x) - \mathbf{P}^T \mathbf{w}(x))^T \boldsymbol{\alpha} \\ &= \mathbf{r}^T \Phi^T \mathbf{q} + (L\mathbf{p}(x) - \mathbf{P}^T \mathbf{w}(x))^T \boldsymbol{\alpha} \end{aligned}$$

$$Ls(x) - L_h s(x) = 0 \text{ due to (17) and (18)} \quad (19)$$

4.3 Consistency Results for Function Approximations

Let u be a solution to (1) and $s \in V_h + \Pi_k$. We can say $u = s + (u - s)$ to get the following:

$$\begin{aligned} \mathcal{E}[u](\mathbf{x}) &:= Lu(\mathbf{x}) - L_h u(\mathbf{x}) \\ &= Lu(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} \\ &= L((u - s) + s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) \mathbf{u} - L_h s(\mathbf{x}) + L_h s(\mathbf{x}) \\ &= L(u - s)(\mathbf{x}) + (Ls(\mathbf{x}) - L_h s(\mathbf{x})) + L_h(s - u)(\mathbf{x}) \\ &= L(u - s)(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) (\mathbf{u} - \mathbf{s}), \text{ by (19)} \\ &= Le(\mathbf{x}) - \mathbf{w}^T(\mathbf{x}) e, e(\mathbf{x}) = u(\mathbf{x}) - s(\mathbf{x}) \\ &= \mathcal{E}[e](\mathbf{x}) \end{aligned}$$

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5 Appendix

Tasks: Define my new RKHS.

Define my reproducing Kernel

→ Simplify to the discrete case
quadrature rule form

→ Show conditions for being
PSD and PD.

→ Show or state is reproducing
property. Convergence guarantees

Discrete RKHS:

→ Smaller subspace for N nodes
instead T

→ Show conditions for nonsingularity
 $\lambda \in \text{range}(\mathcal{Q}^T) \cap \text{Ker}(\mathcal{R}^T)$

→ Scheme degenerates to standard
RK-FD when $X=Y$

→ $\lambda_{\min}(\mathcal{G})$ improves, eigenvalues
increase monotonically.

Final why $\mathcal{G}$ is nondecreasing in the
lower order.

$$\mathcal{H}_K(u_i) \subseteq \mathcal{H}_K(u_{i+1})$$

$\mathcal{G} \mapsto \mathcal{G} + \mathcal{V}\mathcal{V}^T$ monotonically.

→ stable for fixed $\mathcal{Q}$ & $\mathcal{R}$, degrades
for PHS with Expanding $\mathcal{Q}$ & $\mathcal{R}$.

9/21/26

The RKHS starts from functions $\phi \in \mathcal{H}^2$

Then define the kernel in the form of

$$K(x, y) = \int \phi(x-z) \phi(y-z) dz$$

By Moore-Aronszajn, K is a proper kernel for an RKHS,

Theorem: Suppose K is a symmetric, p.d. kernel on a set X . Then there is a unique Hilbert space of functions on X for which K is a reproducing kernel

* Define RKHS explicitly,

Show it is positive Definite on $\mathcal{Y}$:

$$G_{\infty} := [K(y_i, y_j)]$$

$$C^T G_{\infty} C = 0 \Rightarrow \sum_k C_k \phi(y_k - z) = 0 =: F(z)$$

Then $C K = 0$ ~~more~~. This starts from

$$K(x, y) = \int \phi(x-z) \phi(y-z) dz$$

$$C^T G_{\infty} C = \sum_{ij} \bar{C}_i C_j K(x_i, y_j)$$

$$= \int_{\Omega^c} \sum_i \bar{C}_i C_j \phi(y_i - z) \phi(y_j - z) dz$$

$$= \int_{\Omega^c} |\sum_i \bar{C}_i \phi(y_i - z)|^2 dz = \int_{\Omega^c} |F(z)|^2 dz$$

Suppose $d \geq 2$, y_i 's are distinct, ϕ is real analytic on $\mathbb{R}^d \setminus \{y_0\}$,
 Ω^c contains non-empty open subsets.
 F variables $\rightarrow$

$F(z) = \sum_k C_k \phi(y_k - z)$ is real analytic on $D = \mathbb{R}^d \setminus \{y_1, \dots, y_k\}$

$\rightarrow D$ is still connected, then
 $F \equiv 0$ on D . (real-analytic identity theorem).

F is continuous, then
 $F(z) = 0 \forall z \in \Omega^c$.

* Do not need y_i 's so far inside the open ball.

Order: ϕ is continuous, then K_{Ω^c} is p.s.d (1)

And if Mercer's theorem applied, then
 K_{Ω^c} is injective and hence,
 $H_{K_{\Omega^c}} \cong L^2(\Omega^c)$. (2)

And if ϕ is real analytic away from $d \geq 2$,
 then $G_{\Omega^c} \neq 0$ on set of nodes.

Through injectivity, $K_{\Omega^c} \neq 0$, and not
~~matrix~~ augmentation required.

$\rightarrow$ form Diagonal D
 $K_M = \sum_{i=1}^M w_i \phi(x-z_i) \phi(y-z_i)$ via quadrature

$\hookrightarrow$ Show this converges to K

$$K_M \xrightarrow{M \rightarrow \infty} K.$$

Node Subspace: $T_M := \text{span} \{K_M(\cdot, y_k)\}_{k=1}^N$

$$G_M = \tilde{\Phi} D \tilde{\Phi}^T$$

$$\dim(T_M) = \text{rank}(G_M) = \text{rank}(\tilde{\Phi}).$$

$$T_M \neq \mathcal{H}_{K_{\text{osc}}}, \text{ best}$$

$$T_M \rightarrow T_{\infty} \subset \mathcal{H}_{K_{\text{osc}}}$$

$\tilde{\Phi}$ has full row rank and W is invertible
 then $G_M \succeq 0$. From how G_M is defined
 and by $K_M \rightarrow K$, then
 $G_M \rightarrow G_{\infty}$.

By Wey's Eigenvalue inequality,

$$\lambda_{\min}(G_M) \geq \lambda_{\min}(G_{\infty}) - \|G_M - G_{\infty}\|_2 > 0.$$

For $f \in T$, exact reproduction. We do
 have $w_M \rightarrow w_{\infty}$. Then there is
 error $w_M^T(x)f - w_{\infty}^T f$ added to the
 convergence analysis given the new RKHS
 norm. The first term with the infinite
 w_{∞} , turns into the standard
 power function. The second is dependent on
 $M \rightarrow \infty$.

In regards to stability $\lambda_{\min}(G_M) \geq \lambda_{\min}(G_0) - \epsilon_M$

$$\|W_M\|_2 \leq \|G_M^{-1}\|_2 \|b_M\|_2$$

$$\leq \frac{\|b_M\|_2}{\lambda_{\min}(G_M)}$$

$$\leq \frac{\|b_M\|_2}{\lambda(G_0) - \epsilon_M} \quad \text{have } \epsilon_M \sim C_x h_x^q?$$

Although...
nevermind, it does work...

$$\begin{aligned} \phi(x-z)\phi(y-z) &= e^{-\epsilon(\|x-z\|^2 + \|y-z\|^2)} \\ &= e^{-\epsilon(\|x\|^2 - 2\|x\|\|z\| + \|z\|^2 + \|y\|^2 - 2\|z\|\|y\| + \|z\|^2)} \\ &= e^{-\epsilon(\|x\|^2 - 2(\|x\| + \|y\|)\|z\| + \|y\|^2 + 2\|z\|^2)} \\ &= e^{-\epsilon(\|x\|^2 - 2(\|x\| + \|y\| - \|z\|)\|z\| + \|y\|^2)} \\ &= e^{-\epsilon(\|x\|^2 - 2(\|x\| + \|y\| - \|z\|)\|z\| + \|y\|^2)} \end{aligned}$$

$$\text{if } \|z\| = \|y\| \Rightarrow$$

~~approximation~~

$$= e^{-\epsilon\|x-y\|^2}, \text{ the standard method.}$$